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MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY, M14, M16, XM18, XM20, XM22, XM24, M54, METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification For

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-413 — Visual Inspection Guide for Rubber O-Rings

MIL-STD-639 — Surface Roughness Values Preferred for Army Ordnance Materiel

MIL-STD-1167 — Ammunition Data Card

DRAWINGS

MUNITIONS COMMAND

D10520862 — Initiator, Cartridge Actuated, Delay, Metal Parts Assembly

C8846785 — Container, Ammunition, Fiber M459 for Initiators, Cartridge Actuated, Delay

C8846786 — Box, Packing, Ammunition, for Initi-

FSC 1377

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ators, Cartridge Actuated, Delay in Fiber Container M459

IEL–10520862 — Index of Inspection Equipment List for Initiator, Cartridge Actuated, Delay M14, M16, XM18, XM20, XM22, XM24, M54, METAL PARTS ASSEMBLY

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified in the applicable drawings, referenced specifications and the following:

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Mechanical properties. Mechanical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum without deformation or mechanical failure (see 6.1).

3.6 Firing pin release. The firing pin of the metal parts assembly, less safety pin, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin .370 inches, minimum, and .425 inches, maximum.

3.7 Initiator pin security. Application of an axial load of 200 to 220 pounds to the initiator pin of the metal parts assembly shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.8 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled hand tight.

3.9 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour.

3.10 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 3° in either direction when passed in front of the compass at a distance of 6 inches.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawing, referenced specifications and the following:

3.11.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.11.2 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.11.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550.

4.2 Initial production sample. At the beginning of regular production, the supplier shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies and 10 sets of unassembled parts. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used for regular production. All parts and material, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Submission of product.* The product shall be submitted in accordance with MIL–STD–105.

4.3.1.2 *Size of lot.* For the purpose of identification and shipment the shipping lots of metal parts assemblies shall consist of 623 metal parts assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in tables I and II. For an individual defect (single characteristic) the criteria in table II shall be used. All nonconforming material shall be rejected. Inspection for rubber "O" rings shall be in accordance with Standard MIL–STD–413. The Ordnance Standard Finish Blocks prescribed in Standard MIL–STD–639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE I.

		Major		*Minor*	
Lot Size	*Sample Size*	*Accept*	*Reject*	*Accept*	*Reject*
1 – 300	100% Examination	–	–	–	–
301 – 1,300	110	1	2	2	3

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TABLE I (cont).

		Major		Minor	
Lot Size	*Sample Size*	*Accept*	*Reject*	*Accept*	*Reject*
1,301 – 3,200	150	2	3	3	4
3,201 – 8,000	225	3	4	4	5
8,001 – 22,000	300	4	5	5	6

TABLE II.

Total Allowable Defects per Sample Size

Sample Sizes for Major Defects	110	150	225	300
No. of Defects in Class				
2 – 3	1	2	3	4
4 – 7	2	3	4	5
8 – 19	3	4	5	7
20 – 43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects	110	150	225	300
No. of Defects in Class				
2 – 3	2	3	4	5
4 – 7	3	4	5	7
8 – 16	4	5	8	10
17 – 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 *Chamber, Initiator, Dwg C8597928*, covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
	None defined.	
MAJOR		
101.	Pitch diameter of large external thread, min.	Gage
102.	Major diameter of large external thread, min.	Gage
103.	Pitch diameter of small external thread, min.	Gage
104.	Major diameter of small external thread, min.	Gage

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	Defect	*Method of Inspection*
MAJOR		
105.	Pitch diameter of internal thread, max.	Gage
106.	Minor diameter of internal thread, max.	Gage
107.	Concentricity of datum location with pitch diameter of small external thread	Gage
108.	Concentricity of pitch diameter of large thread with cavity	Gage
109.	Concentricity of large external thread pilot with large external thread	Gage
110.	Concentricity of counterbore with cavity including perpendicularity with base of counterbore	Gage
111.	Diameter of counterbore	Gage
112.	Diameter of large thread pilot	Gage
113.	Depth of cavity	Gage
114.	Length to datum location	Gage
115.	Thickness of wall, min.	Gage
116.	Metal defective	Visual
117.	Protective finish, inadequate or defective	Visual
118.	Surface finish of cavity and angle of gas port, improper	Visual
119.	Burr or sharp corner	Visual
120.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of large external thread, max.	Gage
202.	Pitch diameter of small external thread, max.	Gage
203.	Pitch diameter of internal thread, min.	Gage
204.	Diameter of small external thread relief	Gage
205.	Diameter of small external thread pilot	Gage
206.	Diameter of hole in gas port	Gage
207.	Diameter of cavity	Gage
208.	Depth of counterbore	Gage
209.	Length of gas port	Gage
210.	Length of small thread relief	Gage
211.	Length from face of gas port to small thread	Gage
212.	Length of external thread pilot	Gage
213.	Thickness of wall, max.	Gage
214.	Depth of threaded hole	Measure
215.	Diameter of chamfer at internal thread	Measure
216.	Total length	Measure
217.	Length of large external thread	Measure

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	Defect	*Method of Inspection*
MINOR		
218.	Marking missing, incorrect or illegible	Visual
219.	Radius or chamfer, missing or incorrect	Visual
220.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.2 *Cap, Dwg F10520552*, covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of small internal thread, max.	Gage
102.	Minor diameter of small internal thread, max.	Gage
103.	Pitch diameter of large internal thread, max.	Gage
104.	Minor diameter of large internal thread, max.	Gage
105.	Concentricity of pitch diameter of large internal thread with midsection cavity including perpendicularity with base of counterbore of large cavity	Gage
106.	Concentricity of pitch diameter of small internal thread with midsection cavity including perpendicularity with base of counterbore of large cavity	Gage
107.	Concentricity of hole between small cavity and boss cavity with midsection cavity including perpendicularity with base of counterbore of large cavity	Gage
108.	Concentricity of undercut in large cavity with large internal thread	Gage
109.	Concentricity of boss cavity with hole between small cavity and boss cavity	Gage
110.	Concentricity of flange with hole between small cavity and boss cavity	Gage
111.	Concentricity of body, large end with large internal thread	Gage
112.	Concentricity of small cavity with midsection cavity	Gage
113.	Concentricity of thread cavity with large internal thread	Gage
114.	Symmetry of thickness of mounting lugs with body, large end	Gage
115.	Symmetry of length of mounting lugs with body, large end	Gage
116.	Variation of center line of mounting holes with small hole between boss cavity and small cavity	Gage
117.	Diameter of large cavity rear of undercut	Gage
118.	Diameter of small cavity	Gage
119.	Diameter of boss cavity, max.	Gage
120.	Diameter of hole between small cavity and boss cavity, max.	Gage
121.	Diameter of undercut at large internal thread	Gage

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	Defect	*Method of Inspection*
MAJOR		
122.	Diameter of safety pin hole, max.	Gage
123.	Diameter of flange	Gage
124.	Length of effective small interior thread, min.	Gage
125.	Length of small interior thread, max.	Gage
126.	Width of undercut behind large interior thread	Gage
127.	Location of safety pin hole	Gage
128.	Height of boss	Gage
129.	Metal defective	Visual
130.	Protective finish, inadequate or defective	Visual
131.	Burr or sharp corner	Visual
132.	Foreign matter	Visual
MINOR		
201.	Pitch diameter at set screw thread	Gage
202.	Minor diameter of set screw thread	Gage
203.	Pitch diameter of small internal thread, min.	Gage
204.	Pitch diameter of large internal thread, min.	Gage
205.	Diameter of body, small end	Gage
206.	Diameter of body, large end	Gage
207.	Diameter of groove, small end	Gage
208.	Diameter of chamfer, large internal thread	Gage
209.	Diameter of chamfer, small internal thread	Gage
210.	Diameter of midsection cavity	Gage
211.	Diameter of boss cavity, min.	Gage
212.	Diameter of hole between boss cavity and small cavity, max.	Gage
213.	Diameter of safety pin hole, min.	Gage
214.	Diameter of mounting hole	Gage
215.	Depth from base of large cavity rear of undercut to base of midsection cavity	Gage
216.	Depth from base of large cavity rear of undercut to face of internal boss	Gage
217.	Depth to bottom of large cavity rear of undercut	Gage
218.	Depth of large cavity rear of undercut	Gage
219.	Total length	Gage
220.	Length of body, large end	Gage
221.	Width of groove, small end	Gage
222.	Width across flat	Gage

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	Defect	*Method of Inspection*
MINOR		
223.	Thickness of mounting flange	Gage
224.	Location of set screw hole	Gage
225.	Location of groove, small end	Gage
226.	Location of small cavity	Gage
227.	Location of mounting holes	Gage
228.	Location of flats	Gage
229.	Length from rear flange to shoulder	Measure
230.	Width of flange	Measure
231.	Safety pin hole obstructed	Visual
232.	Marking missing, incorrect or illegible	Visual
233.	Radius or chamfer, missing or incorrect	Visual
234.	Surface finish, improper	Visual

4.3.2.3 *Pin, Initiator, Dwg C10520553*, covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
None defined.		
MAJOR		
101.	Concentricity of groove in head with body	Gage
102.	Concentricity of shoulder with body	Gage
103.	Symmetry of pull hole with body	Gage
104.	Diameter of undercut, min., if applicable	Gage
105.	Width of undercut, max., if applicable	Gage
106.	Diameter of body	Gage
107.	Diameter of shoulder	Gage
108.	Diameter of groove in body, min.	Gage
109.	Width of head groove	Gage
110.	Location of groove in head	Gage
111.	Location of groove in body	Gage
112.	Location of pull hole	Gage
113.	Profile of end	Visual
114.	Profile of head	Visual
115.	Metal defective	Visual
116.	Protective finish, inadequate or defective	Visual

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	Defect	*Method of Inspection*
MAJOR		
117.	Surface finish of groove in head and face of shoulder, improper	Visual
118.	Burr	Visual
119.	Foreign matter	Visual
MINOR		
201.	Diameter of groove in head	Gage
202.	Diameter of groove in body	Gage
203.	Diameter of pull hole	Gage
204.	Length to groove in body	Gage
205.	Width of flat at body groove end	Gage
206.	Thickness of pull hole end	Gage
207.	Total length	Measure
208.	Marking missing, incorrect or illegible	Visual
209.	Radius or chamfer, missing or incorrect	Visual
210.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.4 *Seal, Dwg B10520415*, covering a detail of Dwg D10520862

	Defect	*Method of Inspection*
CRITICAL		
None defined		
MAJOR		
101.	Diameter, outside, min.	Gage
102.	Width, min.	Gage
103.	Thickness, min.	Gage
104.	Cut, crack or split	Visual
105.	Flash	Visual
106.	Foreign matter	Visual
MINOR		
201.	Diameter, outside, max.	Gage
202.	Width, max.	Gage
203.	Thickness, max.	Gage

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4.3.2.5 *Housing, Firing Pin, Dwg C10520410*, covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
1.	Diameter of striker hole, min.	Gage
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of striker hole with cavity	Gage
104.	Concentricity of cavity with body	Gage
105.	Concentricity of pitch diameter of thread with body	Gage
106.	Perpendicularity of face with body	Gage
107.	Diameter of body, min.	Gage
108.	Diameter of cavity	Gage
109.	Diameter of flange, min.	Gage
110.	Diameter of boss	Gage
111.	Diameter of undercut, min.	Gage
112.	Total length, max.	Gage
113.	Width of undercut	Gage
114.	Thickness of striker hole	Gage
115.	Location of thread	Gage
116.	Chamfer at striker hole	Gage
117.	Height of boss	Gage
118.	Metal defective	Visual
119.	Protective finish, inadequate or defective	Visual
120.	Surface finish of cavity, improper	Visual
121.	Burr	Visual
122.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of striker hole, max.	Gage
204.	Diameter of flange, max.	Gage
205.	Diameter of wrench holes	Gage
206.	Depth of whench holes	Gage
207.	Total length, min.	Gage
208.	Length of body to flange	Gage

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	Defect	*Method of Inspection*
MINOR		
209.	Location of wrench holes	Gage
210.	Marking missing, incorrect or illegible	Visual
211.	Radius or chamfer, missing or incorrect, except as otherwise classified	Visual
212.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.6 *Pin, Firing, Dwg C10520411,* covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of large cavity with body	Gage
102.	Concentricity of striker with body	Gage
103.	Concentricity of cavity with body	Gage
104.	Symmetry of pin relief holes with small cavity	Gage
105.	Symmetry of ball hole counterbore with ball hole	Gage
106.	Diameter of striker	Gage
107.	Diameter of body	Gage
108.	Diameter of ball hole, max.	Gage
109.	Diameter of ball hole counterbores	Gage
110.	Diameter of pin relief holes, min.	Gage
111.	Depth of ball counterbore, min.	Gage
112.	Length of striker, min.	Gage
113.	Location of ball hole counterbore	Gage
114.	Metal defective	Visual
115.	Protective finish, inadequate or defective	Visual
116.	Surface finish of body, improper	Visual
117.	Burr or sharp corner	Visual
118.	Foreign matter	Visual
MINOR		
201.	Diameter of large cavity	Gage
202.	Diameter of small cavity	Gage
203.	Diameter of ball hole, min.	Gage
204.	Diameter of pin relief holes, max.	Gage

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	Defect	*Method of Inspection*
MINOR		
205.	Depth of large cavity	Gage
206.	Depth to bottom of small cavity	Gage
207.	Depth of ball hole counterbore, max.	Gage
208.	Length of striker, max.	Gage
209.	Total length	Gage
210.	Location of pin relief holes	Gage
211.	Marking missing, incorrect or illegible	Visual
212.	Radius or chamfer, missing or incorrect	Visual
213.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.7 *Spring, Initiator, Dwg B10520412,* covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
101.	Load at specified height	Gage
102.	Closed length, max.	Gage
103.	Ends, incorrect	Visual
104.	Protective finish, inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
201.	Diameter, outside	Gage
202.	Free length	Gage

4.3.2.8 *Pin, Safety. Dwg B10520413,* covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of wire	Gage
102.	Profile, incorrect	Visual

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	Defect	*Method of Inspection*
MAJOR		
103.	Metal defective	Visual
104.	Protective finish, inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
201.	Diameter of head, outside	Measure
202.	Length between high and low ends	Measure
203.	Total length	Measure
204.	Location of center loop	Measure
205.	Radius of center loop, incorrect	Measure

4.3.2.9 *Filter, Dwg C10520414*, covering a detail of Dwg D10520862.

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of thread with center hole and outside diameter	Gage
104.	Symmetry of holes with outside diameter	Gage
105.	Diameter of center hole	Gage
106.	Diameter of holes	Gage
107.	Depth of center hole	Gage
108.	Depth of slot, max.	Gage
109.	Location of holes	Gage
110.	Angular location of holes	Gage
111.	Nylon patch not properly located or fused	Manual/Visual
112.	Number of holes, incorrect	Visual
113.	Metal defective	Visual
114.	Protective finish, inadequate or defective	Visual
115.	Burr	Visual
116.	Foreign matter	Visual

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	Defect	*Method of Inspection*
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of undercut	Gage
203.	Diameter of body	Gage
204.	Depth of slot, min.	Gage
205.	Total length	Gage
206.	Length of body	Gage
207.	Width of undercut, max.	Gage
208.	Width of slot	Gage
209.	Location of slot	Gage
210.	Marking missing, incorrect or illegible	Visual
211.	Chamfer, missing or incorrect	Visual
212.	Surface finish, improper	Visual

4.3.2.10 *Assembly, Dwg D10520862,* less all Dwgs except C10520410, C10520411 and CCAX3–587929.

	Defect	*Method of inspection*
CRITICAL		
1.	Firing pin, protrusion	Gage
2.	Steel ball, missing	Gage
MAJOR		
101.	Burr	Visual
102.	Foreign matter	Visual
MINOR		
	None defined	

4.3.2.11 *Assembly, Dwg D10520862.*

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
101.	Firing pin not cocked	Manual
102.	Initiator pin not free to rotate	Manual
103.	Any part missing, except as otherwise classified	Visual

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	Defect	*Method of Inspection*
MAJOR		
104.	Improper assembly	Visual
105.	Safety pin improperly inserted	Visual
106.	O-ring not lubricated	Visual
107.	Marking missing, incorrect or illegible	Visual
MINOR		
201.	Shipping cap missing	Visual
202.	Protective finish, inadequate or defective	Visual

4.3.2.12 *Container, Fiber, Dwg C8846785.*

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual
202.	Metal ends loose or distorted	Manual
203.	Gap between body and cover, max.	Visual
204.	Cut or damaged through all impregnated layers	Visual

4.3.2.13 *Box, Packing, Dwg C8846786.*

	Defect	*Method of Inspection*
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Manual/Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual/Visual
204.	Contents of box exposed	Visual

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	Defect	*Method of Inspection*
MINOR		
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with 4.4.

4.3.3.1 *Hardness.* Fifty units from each heat treated lot of components shall be tested for compliance with the requirements specified on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 *Hydrostatic pressure.* Each chamber and cap subassembly shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly.

4.3.3.3 *Firing pin release.* The initiator pin of each partial assembly consisting of all parts except the safety pin and chamber shall be subject to the firing pin release test. Failure of any partial assembly to comply with the requirements of 3.6 shall be cause for rejection of the subassembly.

4.3.3.4 *Initiator pin security.* The initiator pin of each subassembly consisting of all parts except the chamber, shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.7 shall be cause for rejection of the subassembly.

4.3.3.5 *Protective finish.* The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of 3.9 and the applicable specification, shall be cause for rejection of the lot.

4.3.3.6 *Residual magnetism.* Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.10 shall be cause for rejection of the assembly.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–10520862.

4.4 Test methods and procedures.

4.4.1 *Hardness.* Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.4.2 *Hydrostatic pressure.* Prior to the application of protective finish, the chamber and cap subassembly shall be mounted in the device, using the fixtures for properly sealing all openings, and subjected to the hydrostatic pressure to determine compliance with 3.5.

4.4.3 *Firing pin release.* The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.4.4 *Initiator pin security.* The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which

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registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.4.5 *Protective finish.* Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.6 *Residual magnetism.* The assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the assembly 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading as shown in figure 1. The assembly shall be moved slowly past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90° rotation of the assembly.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking—Level A. Packing and marking shall be in accordance with drawings C8846785 and C8846786.

5.2 Data cards. Data card information shall be as specified in MIL–STD–1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurence of which is other than a design function.

Custodian:

Army—MU

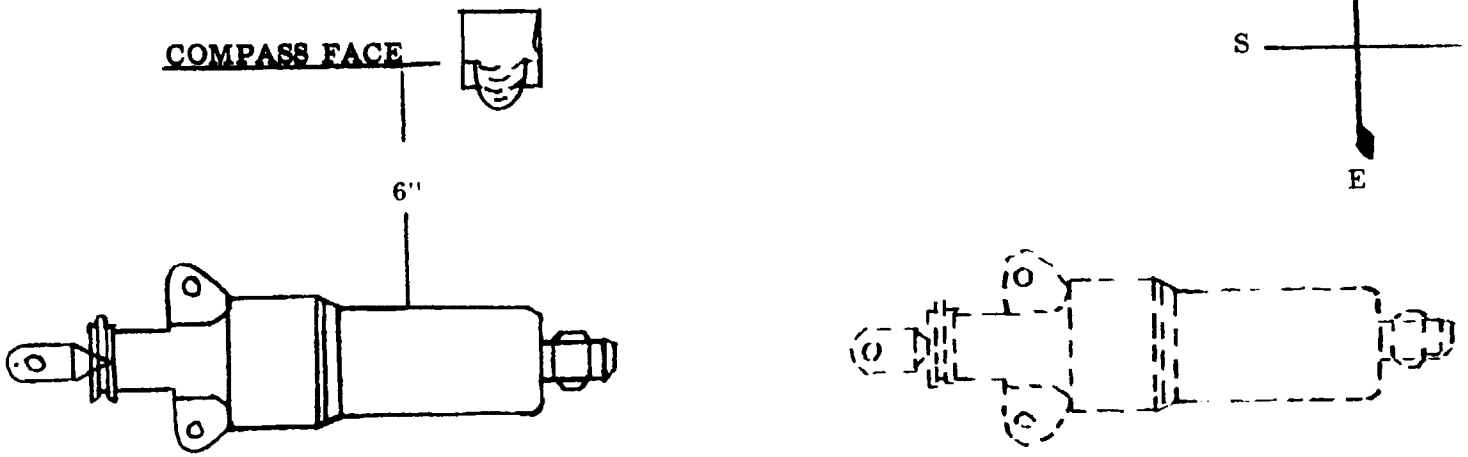
Preparing activity:

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Project No. 1377–A045

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PILOT'S STANDBY COMPASS, TYPE AN-5766



MOVE FROM THIS POSITION

TO THIS POSITION

PLAN VIEW

NOTE:

INITIATOR AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN INITIATOR, CARTRIDGE ACTUATED, DELAY, M14,M16,XM-18,XM20, XM22, XM24, M54

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

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